

Module 24.05: Underfitting (High Bias & Oversimplification)

1. Theoretical Definition of Underfitting

Underfitting occurs when a machine learning model is overly simplistic and lacks the expressive mathematical capacity to capture the underlying structure, curvature, and relationships present in the data.

Target Ground Truth: Non-linear parabolic curve ($y = 2x^2 + 3$)

Underfitted Model: Simple straight line ($y = wx + b$)

Result:

- * Training Error -> High (Fails even on the data it was trained on)
- * Test/Val Error -> High (Equally poor generalization)

2. The Bias-Variance Perspective: High Bias

Underfitting represents the **High Bias** regime of the Bias-Variance tradeoff:

$$\text{Bias} \left[\hat{f}(x) \right] = \mathbb{E} \left[\hat{f}(x) \right] - f(x) \gg 0$$

- The fundamental assumptions made by the model family are too restrictive.
 - For instance, forcing a strictly linear hypothesis $\hat{y} = wx + b$ onto an inherently sinusoidal or quadratic phenomenon.
 - Even if infinite training samples ($m \rightarrow \infty$) were provided, an underfitted model cannot improve beyond its rigid capacity boundary.
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3. Causes and Diagnostic Signatures of Underfitting

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| Root Causes of Underfitting
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| 1. Model Too Simple: Linear model applied to non-linear physical systems.
| 2. Insufficient / Uninformative Features: Critical predictive signals are missi
| 3. Excessive Regularization: Hyperparameter lambda is set so large that weights
| 4. Under-trained Model: Gradient descent terminated before reaching local/globa
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Signature Diagnostic Sign:

Both training and validation loss remain unacceptably high with minimal gap:

$$\begin{aligned} \mathcal{L}_{\text{train}} &\approx \text{High}, \quad \mathcal{L}_{\text{val}} \approx \text{High}, \\ &\quad (\mathcal{L}_{\text{val}} - \mathcal{L}_{\text{train}} \approx 0) \end{aligned}$$